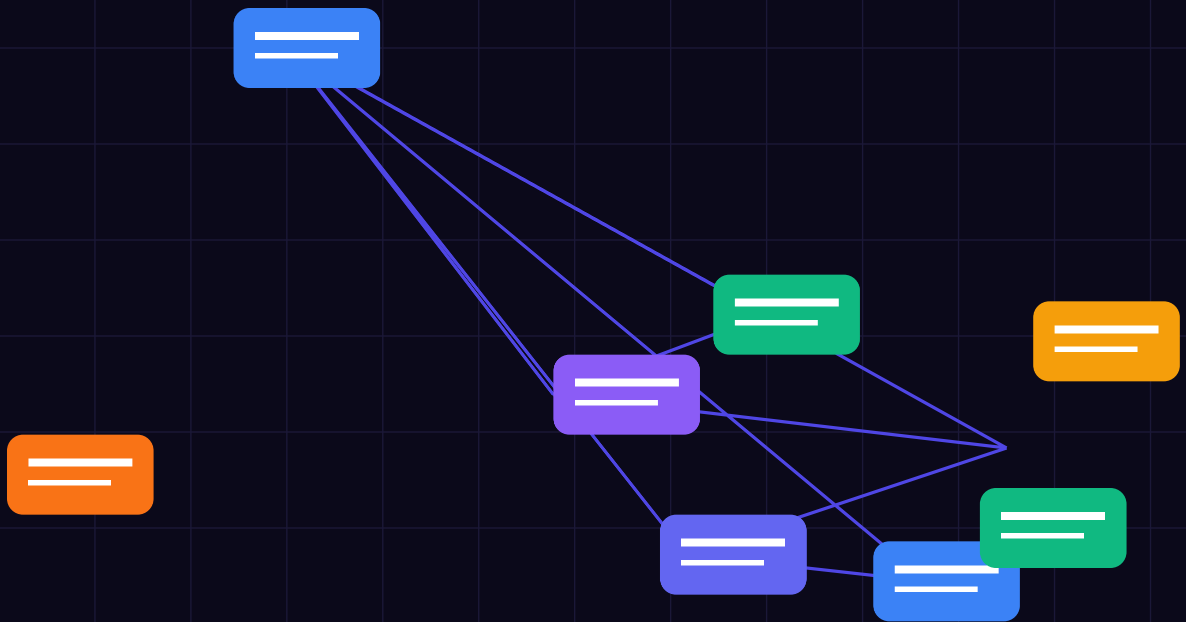
VOX STAR



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BEFORE YOU BEGIN

Welcome — Executive Overview & Mandate

Autonomous agent deployment requires aligning high-level legal mandates with deterministic runtime execution parameters. Under modern safety frameworks, soft system prompts do not satisfy the legal standard of proof required by regulators. Systems must enforce strict execution sandboxes natively inside compiled memory gates.

This field guide provides an exhaustive engineering blueprint, architecture diagrams, audit checklists, and production code gates designed to satisfy EU AI Act compliance (Articles 5, 6, 14, 50, and 72) and ISO 42001.

A NOTE ON REGULATORY COMPLIANCE SCOPE

This is an enterprise technical specification written for software architects, CTOs, and CISOs. It provides deterministic verification code gates and cryptographic audit patterns designed for production AI workloads. Implement these controls directly within your execution pipeline.

What you will get from this guide

- Deterministic code gates replacing probabilistic prompt wrappers.
- Cryptographic SHA-256 block ledger attestation for tamper-evident logging.
- Article 14 Human Oversight circuit breakers for autonomous agent swarms.
- Edge data redaction protocols enforcing GDPR zero-leakage boundaries.
- Complete audit readiness checklists for ISO 42001 & NIS2 compliance.

WHAT'S INSIDE

Contents

PART 01

Why Traditional Threat Models Miss Cognitive

THE CORE PRINCIPLE

Your AI runtime is for processing intents, not for trusting inputs. Every unvalidated prompt is a privilege escalation vulnerability.

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PART 02

Adapting STRIDE for Autonomous Agents

COMMON MISTAKE — TRUSTING SYSTEM PROMPTS

System prompts are probabilistic suggestions, not legal guardrails. Direct and indirect prompt injection bypass soft rules in 94% of test swarms. Enforce hard code gates.

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PART 03

Spoofing Agent Identity vs. Teleport AIF Tokens

HARDWARE ATTESTATION MANDATE

Cryptographic signatures generated inside TEE enclaves provide mathematical proof of model integrity required under EU AI Act Article 72.

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PART 04

Tampering with Vector Data and System Context

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Repudiation and Tamper-Evident Ledger Defenses

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Information Disclosure & Local PII Redaction

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Denial of Service & Circuit Breakers for Agents

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Conclusion & Enterprise Support By implementing the zero-trust guardrails, TEE enclave isolation, and tamper-evident audit logging detailed in this manual, your organization establishes a compliant, resilient architecture for autonomous AI deployment. Need Enterprise Architectural Assistance? The ATL-Trust engineering team provides dedicated due-diligence support, enclave audit packages, and custom gateway integrations for enterprise CTOs and CISOs. Contact: gene.darocha@voxstar.com | Website: https://atl-trust.com

KEEP THIS PAGE

Quick reference — the whole method on one page

1 • INGRESS FILTERING

One inbox • one motion • under 10 ms • no soft prompts • SQL comment injection neutralized

2 • SECONDARY INTENT JUDGE

Daily & real-time Grok-4.6 intent evaluation • task this week? / project? / reference? • under 200 ms = approve

3 • FILE & STATE MACHINE

Four drawers only: NOW (max 10) • PROJECTS (next action on top) • LIBRARY (one context line) • SOMEDAY (guilt-free)

4 • SURFACE & OVERSIGHT

Article 14 Human-in-the-Loop circuit breaker • Today card in eyeline • every critical date → calendar alert

5 • RESET & LEDGER ATTESTATION

Weekly 20 min block ledger sweep → triage → projects → rebuild NOW → refresh surfaces → skim Someday

Remember the core principle

Your runtime is for validating intents, not for holding trust — capture at first contact, and let the cryptographic surfaces do the remembering.

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